

Practical -07

Code:

```
GNU nano 7.2
#include <stdio.h>
#include <stdlib.h>

int global_var = 10;
static int static_var = 20;

void code_function()
{
}

int main()
{
    int stack_var = 30;

    int *heap_var = malloc(sizeof(int));
    *heap_var = 40;

    printf("Code    : %p\n", (void *)code_function);
    printf("Global  : %p\n", (void *)&global_var);
    printf("Static  : %p\n", (void *)&static_var);
    printf("Heap    : %p\n", (void *)heap_var);
    printf("Stack   : %p\n", (void *)&stack_var);

    getchar();

    free(heap_var);

    return 0;
}
```

Part -1

Output:

```
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-07$ nano memory.c
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-07$ gcc memory.c -o memory
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-07$ ./memory
Code    : 0x566718da91c9
Global  : 0x566718dac010
Static  : 0x566718dac014
Heap    : 0x5667205b82a0
Stack   : 0x7fff411713dc
```

Part -2

Output:

```

bharath_sai_reddy@LAPTOP-1711DF88: ~/90128_OSSP/2520090128_Practical/Practical-07$ ps -ef | grep ./memory
bharath+ 558 311 0 09:38 pts/0 00:00:00 ./memory
bharath+ 614 569 0 09:47 pts/2 00:00:00 grep --color=auto ./memory
bharath_sai_reddy@LAPTOP-1711DF88: ~/90128_OSSP/2520090128_Practical/Practical-07$ cat /proc/558/maps
566718da8000-566718da9000 r--p 00000000 08:30 19618 /home/bharath_sai_reddy/90128_OSSP/2520090128_Practical/Practical-07/memory
566718da9000-566718daa000 r-xp 00001000 08:30 19618 /home/bharath_sai_reddy/90128_OSSP/2520090128_Practical/Practical-07/memory
566718daa000-566718dab000 r--p 00002000 08:30 19618 /home/bharath_sai_reddy/90128_OSSP/2520090128_Practical/Practical-07/memory
566718dab000-566718dac000 r--p 00002000 08:30 19618 /home/bharath_sai_reddy/90128_OSSP/2520090128_Practical/Practical-07/memory
566718dac000-566718dad000 rw-p 00003000 08:30 19618 /home/bharath_sai_reddy/90128_OSSP/2520090128_Practical/Practical-07/memory
5667205b8000-5667205d9000 rw-p 00000000 00:00 0 [heap]
7c9c9ea00000-7c9c9ea28000 r--p 00000000 08:30 11972 /usr/lib/x86_64-linux-gnu/libc.so.6
7c9c9ea28000-7c9c9ebb1000 r-xp 00028000 08:30 11972 /usr/lib/x86_64-linux-gnu/libc.so.6
7c9c9ebb1000-7c9c9ec00000 r--p 001b1000 08:30 11972 /usr/lib/x86_64-linux-gnu/libc.so.6
7c9c9ec00000-7c9c9ec04000 r--p 001ff000 08:30 11972 /usr/lib/x86_64-linux-gnu/libc.so.6
7c9c9ec04000-7c9c9ec06000 rw-p 00203000 08:30 11972 /usr/lib/x86_64-linux-gnu/libc.so.6
7c9c9ec06000-7c9c9ec13000 rw-p 00000000 00:00 0
7c9c9ec42000-7c9c9ec45000 rw-p 00000000 00:00 0
7c9c9ec4c000-7c9c9ec4e000 rw-p 00000000 00:00 0
7c9c9ec4e000-7c9c9ec52000 r--p 00000000 00:00 0 [vvar]
7c9c9ec52000-7c9c9ec54000 r--p 00000000 00:00 0 [vvar_vclock]
7c9c9ec54000-7c9c9ec56000 r-xp 00000000 00:00 0 [vdso]
7c9c9ec56000-7c9c9ec57000 r--p 00000000 08:30 11967 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7c9c9ec57000-7c9c9ec82000 r-xp 00001000 08:30 11967 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7c9c9ec82000-7c9c9ec8c000 r--p 0002c000 08:30 11967 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7c9c9ec8c000-7c9c9ec8e000 r--p 00036000 08:30 11967 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7c9c9ec8e000-7c9c9ec90000 rw-p 00038000 08:30 11967 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7fff41153000-7fff41174000 rw-p 00000000 00:00 0 [stack]
bharath_sai_reddy@LAPTOP-1711DF88: ~/90128_OSSP/2520090128_Practical/Practical-07$ grep heap /proc/558/maps
5667205b8000-5667205d9000 rw-p 00000000 00:00 0 [heap]
bharath_sai_reddy@LAPTOP-1711DF88: ~/90128_OSSP/2520090128_Practical/Practical-07$ grep stack /proc/558/maps
7fff41153000-7fff41174000 rw-p 00000000 00:00 0 [stack]
bharath_sai_reddy@LAPTOP-1711DF88: ~/90128_OSSP/2520090128_Practical/Practical-07$ █

```